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Addition code:

```
Addition.py > ...
1  num1 = float(input("Enter first number: "))
2  num2 = float(input("Enter second number: "))
3
4  result = num1 + num2
5
6  print("Addition is:", result)
7
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v [icon] [icon] [icon] [icon] X

PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/Addition.py"
Enter first number: 5
Enter second number: 3
Addition is: 8.0
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```

Subtraction Code :

```
subtraction.py > ...
1  # Subtraction of two numbers
2
3  num1 = float(input("Enter first number: "))
4  num2 = float(input("Enter second number: "))
5
6  result = num1 - num2
7
8  print("Result:", result)
9
```

Multiplication Code :

```
multiplication.py > ...
1  num1 = float(input("Enter first number: "))
2  num2 = float(input("Enter second number: "))
3
4  result = num1 * num2
5
6  print("Result:", result)
7
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v [icon] [icon] [icon] [icon] [icon] [icon] [icon] [icon]
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/multiplication.py"
Enter first number: 5
Enter second number: 4
Result: 20.0
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```

Division Code :

```
division.py > ...
1  num1 = float(input("Enter numerator: "))
2  num2 = float(input("Enter denominator: "))
3
4  if num2 != 0:
5      result = num1 / num2
6      print("Result:", result)
7  else:
8      print("Error: Division by zero is not allowed.")
9
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v [icon] [icon] [icon] [icon] [icon] [icon] [icon] [icon]
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/division.py"
Enter numerator: 50
Enter denominator: 20
Result: 2.5
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```

Question 1 :

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula: $p = m \times v$ where: m is the mass of the object (in kilograms). v is the velocity of the object (in meters per second).

Input Format: A single floating-point number representing the mass of the object in kilograms. A single floating-point number representing the velocity of the object in meters per second. Output Format: The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Code :

```
momentum.py > ...
1 # Get inputs from the user
2 mass = float(input("Enter mass in kg: "))
3 velocity = float(input("Enter velocity in m/s: "))
4
5 # Calculate momentum
6 momentum = mass * velocity
7
8 # Display the result rounded to 2 decimal places
9 print(f"The momentum is {momentum:.2f} kgm/s")
```

Output :

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python +v [] [] ... | [] X
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3
.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/momentum.py"
Enter mass in kg: 20.35
Enter velocity in m/s: 12.30
The momentum is 250.31 kgm/s
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```

Question 2:

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Write a Python program that accepts an integer n as input. Depending on the number of digits in n . Constraints: $1 \leq n \leq 999$

Input Format: The input consists of a single integer n . Output Format: If n is a single-digit number, print its square. If n is a two-digit number, print its square root (rounded to two decimal places). If n is a three-digit number, print its cube root (rounded to two decimal places). Else print "Invalid".

Code :

```
n number code.py > ...
1  import math
2
3  n = int(input())
4
5  if 1 <= n <= 999:
6      digits = len(str(n))
7
8      if digits == 1:
9          print(n ** 2)
10
11     elif digits == 2:
12         print(f"{math.sqrt(n):.2f}")
13
14     elif digits == 3:
15         print(f"{n ** (1/3):.2f}")
16
17 else:
18     print("Invalid")
```

Output :

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python: n number code + - [ ] ... [ ] X

PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/n number code.py"
2
4
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/n number code.py"
21
4.58
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class>
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/n number code.py"
584
8.36
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/n number code.py"
5555555
Invalid
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```

Question 3:

Write a Python program that reads the birth date and salary of employees.

Input Format:
The input consists of:
A string representing the birth date of the employee in the format *DD - MM - YYYY*.
A floating-point number representing the salary of the employee in rupees.

Output Format:
The output should include:
The age of the employee.
The salary of the employee in dollars.

Note:
1INR=0.012USD

Code :

```

birth_date_and_salary.py > ...
1  from datetime import datetime
2
3  # Input birth date
4  birth_date = input("Enter birth date (DD-MM-YYYY): ")
5
6  # Input salary in rupees
7  salary_inr = float(input("Enter salary in rupees: "))
8
9  # Convert birth date string to date object
10 birth_date_obj = datetime.strptime(birth_date, "%d-%m-%Y")
11
12 # Get current date
13 today = datetime.today()
14
15 # Calculate age
16 age = today.year - birth_date_obj.year
17
18 # Adjust age if birthday hasn't occurred yet this year
19 if (today.month, today.day) < (birth_date_obj.month, birth_date_obj.day):
20     age -= 1
21
22 # Convert salary to dollars
23 salary_usd = salary_inr * 0.012
24
25 # Output
26 print("Age of employee:", age)
27 print("Salary in dollars:", salary_usd)

```

Output :

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python: birth_date_and_salary + v [ ] [ ] ... [ ] [ ] X

Enter salary in rupees: 50210
Age of employee: 19
Salary in dollars: 602.52
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class>

```

Question 4 :

You are given an integer number. Your task is to reverse the digits of the number and print the reversed number.

Input Format

The input is an integer.

Output Format

Print a single integer which is the reversed number.

Code :

```
reverse_digit.py > ...
1  num = int(input("Enter a number: "))
2
3  reverse = 0
4  while num != 0:
5      digit = num % 10
6      reverse = reverse * 10 + digit
7      num = num // 10
8
9  print(reverse)
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python: reverse_digit + v [ ] [ ] ... [ ] [ ] X

PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/reverse_digit.py"
Enter a number: 12345
54321
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```


Question 5:

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number .

Code :

```
table.py > ...
1  # Take integer input from user
2  num = int(input())
3
4  # Print multiplication table from 1 to 10
5  for i in range(1, 11):
6      print(num, "x", i, "=", num * i)
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python: table + v [ ] [ ] ... | [ ] [ ] x
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> & "C:/Users/ADITYA PATHADE/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/ADITYA PATHADE/OneDrive/Desktop/python class/table.py"
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
PS C:\Users\ADITYA PATHADE\OneDrive\Desktop\python class> |
```